

Al-Amin Hossain Nahid

AI & Full-Stack Developer | CS Undergraduate

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LinkedIn: Al Amin Hossain Nahid | **GitHub:** AlAminNahid | **Portfolio:** alamin-nahid.is-a.dev

Summary

Computer Science undergraduate at the American International University–Bangladesh (AIUB) building LLM-powered systems alongside full-stack web and Android applications. Experienced in TypeScript, Next.js, Python, FastAPI, NestJS, relational data modelling, authenticated REST APIs, and end-to-end feature development. Currently focused on applied LLM engineering, including multi-step agentic workflows, tool calling, retrieval over private document collections, streaming inference, and document analysis. Research interests include machine learning, large language models, and image processing.

Skills

- **Programming Languages:** C/C++, Java, C#, Python, JavaScript, TypeScript, Kotlin
- **Frontend:** HTML, CSS, Tailwind CSS, shadcn/ui, React, React Native, Next.js
- **Backend:** NestJS, FastAPI, ASP.NET, PHP
- **Databases & ORMs:** PostgreSQL, MySQL, MongoDB, TypeORM, Prisma, SQLAlchemy, Alembic
- **Mobile:** Android SDK, Retrofit, MVVM, Material Design
- **AI/LLM:** OpenAI SDK, OpenRouter, Agent Harness, Tool Calling, Agentic Workflows, RAG, Structured Extraction, SSE Streaming
- **Tools:** Git, Docker, Postman, Linux, Android Studio, Firebase, Vercel, Render, Railway

Experience

AuraDev LTD | Software Engineering Intern

June 2026 – Present

- Contributing to **GrantOS**, an AI-assisted grant management platform, across a Next.js/React frontend and asynchronous FastAPI backend, with primary ownership of the agent layer.
- Built a tool-driven agent system that takes users from raw grant sources to drafted proposals through multi-step workflows with defined human approval gates.
- Designed persistent agent state and conversation management, including automatic context compaction, server-driven workflow phases, and per-call token and cost tracking.
- Implemented retrieval over organisational knowledge bases using metadata-based document ranking and LLM-assisted selection, with adaptive document retrieval and context distillation.
- Developed authenticated REST APIs and domain logic using FastAPI, async SQLAlchemy, Alembic, and PostgreSQL, with Clerk JWT authentication and organisation-based access control.
- Built document workflows for PDF/DOCX processing and Cloudflare R2 file storage, alongside proposal version history and collaborative commenting with Plate.js.
- **Technologies:** TypeScript, React, Next.js, Zustand, Plate.js, Python, FastAPI, PostgreSQL, SQLAlchemy, Alembic, Clerk, OpenAI SDK, OpenRouter, Cloudflare R2

Projects

Bilimbi Studio | Frontend Developer

Live · GitHub

- Built and deployed a production marketing website for Bilimbi Studio, showcasing its projects, team, development process, and contact functionality.
- Developed a scalable design-token system with Tailwind CSS v4 to maintain consistent branding and styling across the application.
- Structured the website as a content-driven system using typed constants, separating content from UI components for easier maintenance and updates.
- Implemented a Next.js API route for the contact form with Zod-based validation and external email service integration.
- **Technologies:** TypeScript, Next.js, React, Tailwind CSS, Zod

AI Chat | Full-Stack AI Developer

Live · GitHub

- Developed a full-stack AI chat application with persistent multi-conversation history using Next.js, Prisma, and PostgreSQL.
- Integrated the Google Gemini API through server-side Next.js route handlers with streamed AI responses, keeping API credentials secure on the server.
- Implemented conversation management with validated REST APIs, typed error handling, and reusable React components and hooks.
- Built a responsive chat interface supporting formatted Markdown responses, including code blocks and lists.
- **Technologies:** TypeScript, Next.js, React, Tailwind CSS, Prisma, PostgreSQL, Google Gemini API

Jotion | Notion-Style Workspace

Live · GitHub

- Building a Notion-inspired document workspace with Next.js, React and TypeScript.
- Integrated Clerk authentication with route protection via Next.js middleware and wired Convex as the real-time backend through a custom Convex + Clerk provider.
- Developed a responsive marketing landing page with a component-driven structure and system-aware light/dark theming using next-themes.
- Established the UI foundation with Tailwind CSS v4, shadcn/Base UI components, and Geist/Inter typography.
- **Technologies:** TypeScript, Next.js, React, Convex, Clerk, Tailwind CSS

MessMaster | Android Developer

GitHub · *Google Play (Internal Testing)*

- Developed a native Android application for shared-living management, covering meal tracking, expense management, utility billing, notices, and member communication.
- Implemented role-based experiences for managers and members, integrating with a RESTful backend using Retrofit and OkHttp with JWT-based authentication.
- Applied MVVM architecture with ViewModel and Kotlin Coroutines for asynchronous operations, with centralized API error handling across repositories.
- Built Material Design interfaces using Kotlin and XML, including dashboards, meal management, utility tracking, notices, profile management, and invite workflows.
- **Technologies:** Kotlin, XML, Android SDK, MVVM, Retrofit, OkHttp, Coroutines, Material Design

MessMasterApp_Backend | Backend Developer

GitHub

- Built and deployed a modular RESTful API for a smart mess management system using NestJS and PostgreSQL, consumed by an Android client.
- Implemented secure authentication and authorization using JWT access/refresh tokens, Passport.js, bcrypt, and role-based access control.
- Developed core modules for mess management, meal tracking, expenses, utility costs, billing, statistics, and notifications.
- Designed a normalized PostgreSQL schema with TypeORM migrations and class-validator DTOs for data integrity and API validation.
- **Technologies:** TypeScript, NestJS, PostgreSQL, TypeORM, JWT, Passport.js, Docker

Publications & Research

Peer-Reviewed Publication

- **Al Amin Hossain Nahid**, Ferdus Hossain, Tariful Islam Fahim, and Salman Zzoha. “Intelligent Waste Classification for Sustainable Urban Development: A MobileNetV3-Based Deep Learning Approach Using the BDWaste Dataset.” In *Proceedings of the International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII 2026)*, 2026. Published in IEEE Xplore.

Education

American International University-Bangladesh
B.Sc. in Computer Science and Engineering

Sept 2022 – Present
CGPA: 3.64/4.00